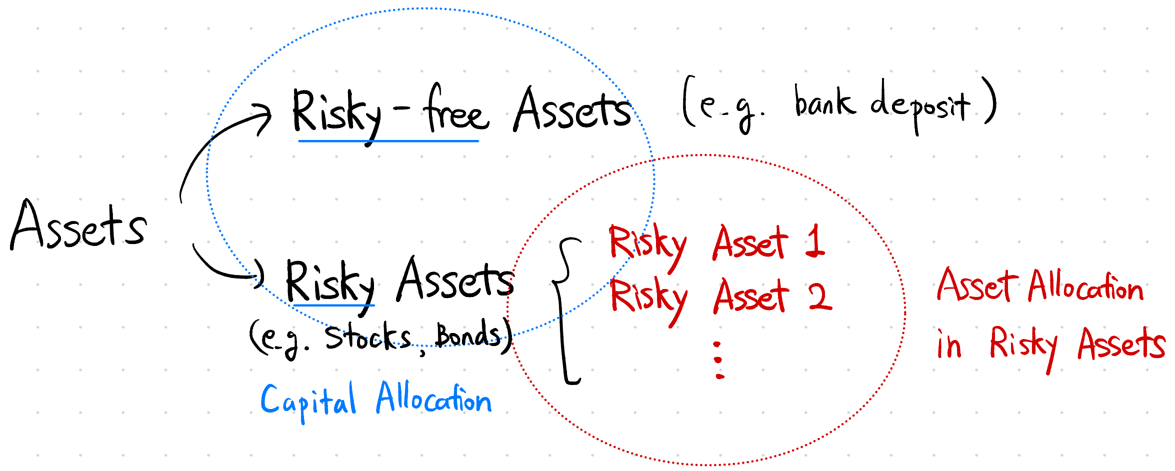
Capital Allocation (Risky Assets)

자본 배분 (Capital Allocation)

위험 자산과 무위험 자산 간의 선택
자산 배분의 기본적인 형태

자산 배분 (Asset Allocation)

광범위한 자산 유형 간의 포트폴리오 선택
(e.g.) 주식, 채권, 안전자산 등 광범위한 자산 유형에 투자하는 비율을 조정하는 것



Capital Allocation Line (CAL, 자본배분선)

투자자에게 주어진 투자 가능한 리스크-수익률 조합을 나타내는 선

- y : 전체 자산 중 위험자산 투자 비중 ($y=1$ implies 100%)
- $1-y$: 전체 자산 중 무위험자산 투자 비중
- $E(r_s)$: 위험 자산의 기대수익률
- σ_s : 위험 자산의 표준편차
- r_f : 무위험 자산의 기대수익률
- $E(r_p)$: 포트폴리오의 기대수익률
- σ_p : 포트폴리오의 표준편차

Example:

Assume that

$r_f = 3\%$, $\sigma_f = 0\%$, $E(r_s) = 10\%$, $\sigma_s = 25\%$

Since $r_p = y \cdot r_s + (1-y) \cdot r_f$,

$$\Rightarrow E(r_p) = y \cdot E(r_s) + (1-y) \cdot r_f$$

(∵) $E(r_f) = r_f$, Since the return of risky-free asset doesn't have volatility, so it is constant. Thus, $E(r_f) = r_f$.

$$= 10y + 3(1-y) = 7y + 3.$$ (Expected Return of Portfolio)

Recall that

$$Var(aX + bY) = a^2 Var(X) + b^2 Var(Y) + 2ab\, Cov(X, Y)$$

for r.v. X and Y.

Since $r_p = y \cdot r_s + (1-y) \cdot r_f$,

$$Var(r_p) = y^2 Var(r_s) + (1-y)^2 Var(r_f) + 2y(1-y) Cov(r_s, r_f)$$

Recall that

$$Cov(X, Y) = E[(X - E[X])(Y - E[Y])]$$

(Actually, if one r.v. has zero std. dev., the Cov. = 0)

$$\Rightarrow Cov(r_s, r_f) = E[(r_s - E(r_s)) \times 0] = E[0] = 0.$$

Then,

$$Var(r_p) = y^2 Var(r_s) + (1-y)^2 Var(r_f) + \cancel{2y(1-y) Cov(r_s, r_f)} \to 0$$

and we've assumed $\sigma_f = 0\%$, so the 2nd term is also zero.

Thus, $\sigma_p = y\sigma_s = 25y$

$$\Leftrightarrow y = \frac{\sigma_p}{25}$$

(Generally, the formula for calculating the variance (risk) of a portfolio consisting of two assets is:

$$\sigma_p^2 = \underbrace{w_s^2}_{\text{weights}} \sigma_s^2 + \underbrace{w_f^2}_{\text{weights}} \sigma_f^2 + \underbrace{2 w_s w_f Cov(r_s, r_f)}_{\text{this term is "Diversification".}}$$

)

$Cov(r_s, r_f)$ could be represented as $\rho_{sf} \sigma_s \sigma_f$

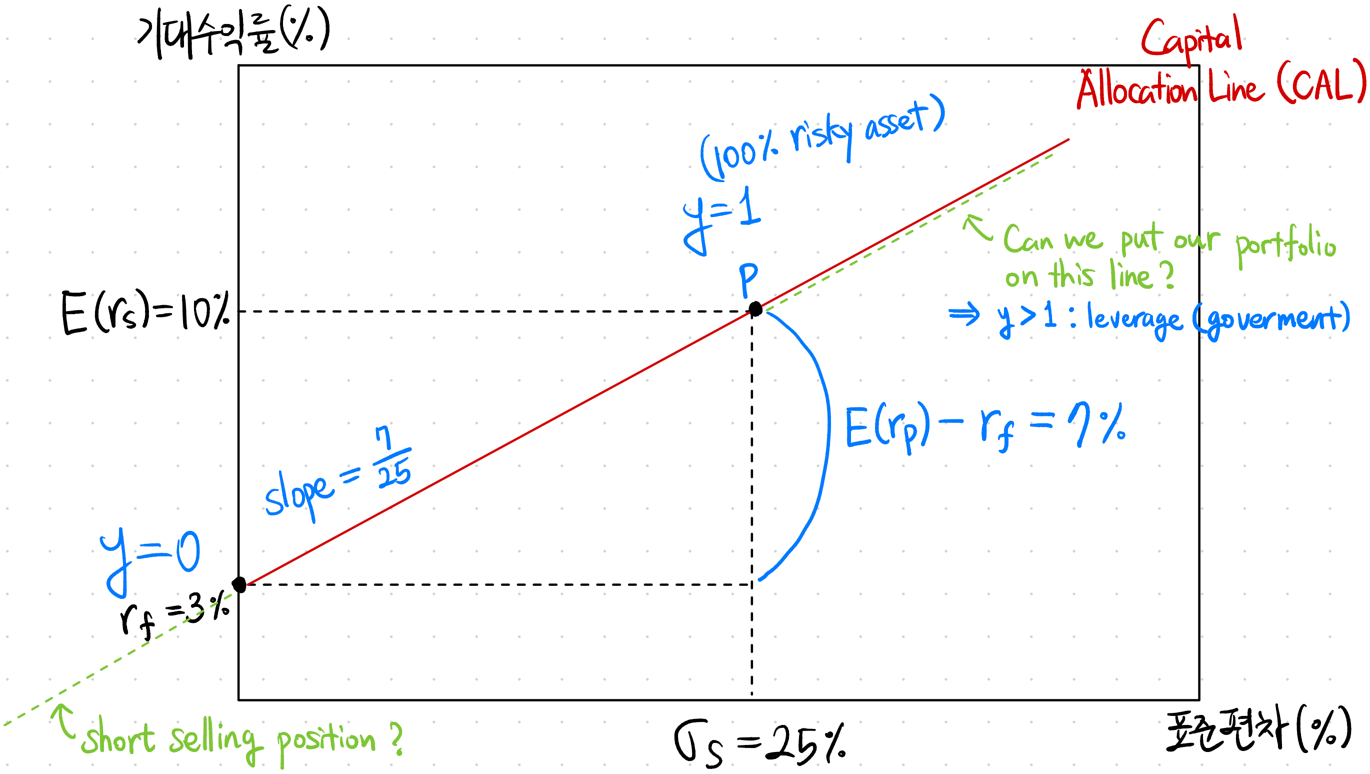
⇒ The smaller the corr. coef. (ρ_{sf}) between two assets (i.e., the more they move differently), the lower the overall risk (σ_p^2) of the portfolio. This is the principle of diversification.

③

Thus, the capital allocation line (CAL) is $E(r_p) = 3 + \frac{7}{25}\sigma_s$.

Actually, $E(r_p) = r_f + \frac{E(r_s) - r_f}{\sigma_s}\sigma_p$.

the numerator $E(r_s) - r_f$ is the excess return.



Investors should take an optimal portfolio with CAL.

Goal : maximize the utility.

$$\max_y U = \max_y \left[E(r_p) - \frac{1}{2}A\sigma_p^2\right]$$
$$= \max_y \left[r_f + y(E(r_s) - r_f) - \frac{1}{2}Ay^2\sigma_s^2\right]$$

Note that above expression is a quadratic function w.r.t. y.

$$\Rightarrow (E(r_s) - r_f) - A\sigma_s^2 y^* = 0$$

$$\Rightarrow y^* = \frac{E(r_s) - r_f}{A\sigma_s^2}$$